

Section B – API Endpoint Plan

The following table defines the RESTful API endpoints planned for the RaceDay system.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as an Organiser or Participant.	None	{ firstName, lastName, email, password, roleId }	201 Created – user registered successfully. 400 Bad Request – invalid or incomplete data. 409 Conflict – email already exists.
POST	/api/auth/login	Authenticates a user and returns an access token.	None	{ email, password }	200 OK – login successful with token. 401 Unauthorized – invalid email or password.
GET	/api/users/profile	Returns the profile details of the currently logged-in user.	Any	None	200 OK – user profile returned. 401 Unauthorized – authentication required. 404 Not Found – user profile does not exist.
PUT	/api/users/profile	Updates the profile details of the currently logged-in user.	Any	{ firstName, lastName, email }	200 OK – profile updated. 400 Bad Request – invalid data. 401 Unauthorized – authentication required. 409 Conflict – email already exists.
GET	/api/events	Retrieves all available events.	None	None	200 OK – list of events returned.
GET	/api/events/{id}	Retrieves details of a specific event.	None	None	200 OK – event returned. 404 Not Found – event does not exist.

POST	/api/events	Creates a new race event.	Organiser	{ eventName, eventDate, location }	201 Created – event created. 400 Bad Request – invalid data. 401 Unauthorized – authentication required. 403 Forbidden – user is not an Organiser.
PUT	/api/events/{id}	Updates an existing race event.	Organiser	{ eventName, eventDate, location }	200 OK – event updated. 400 Bad Request – invalid data. 403 Forbidden – user is not the event Organiser. 404 Not Found – event does not exist.
DELETE	/api/events/{id}	Deletes an existing race event.	Organiser	None	204 No Content – event deleted. 403 Forbidden – user is not authorised. 404 Not Found – event does not exist.
GET	/api/categories	Retrieves all race categories.	None	None	200 OK – list of categories returned.
GET	/api/categories/{id}	Retrieves a specific race category.	None	None	200 OK – category returned. 404 Not Found – category does not exist.
POST	/api/categories	Creates a new race category.	Organiser	{ categoryName }	201 Created – category created. 400 Bad Request – invalid data. 403 Forbidden – user is not an Organiser.
PUT	/api/categories/{id}	Updates an existing race category.	Organiser	{ categoryName }	200 OK – category updated. 400 Bad Request – invalid data. 404 Not Found – category does not exist.
DELETE	/api/categories/{id}	Deletes an existing race category.	Organiser	None	204 No Content – category deleted. 404 Not Found –

					category does not exist.
GET	/api/events/{eventId}/enrolments	Retrieves all enrolments for a specific event.	Organiser	None	200 OK – enrolment list returned. 404 Not Found – event does not exist.
POST	/api/events/{eventId}/enrolments	Enrols the logged-in participant in an event category.	Participant	{ eventCategoryId }	201 Created – enrolment created. 400 Bad Request – invalid category. 401 Unauthorized – authentication required. 409 Conflict – participant already enrolled.
DELETE	/api/enrolments/{id}	Cancels an existing enrolment.	Participant	None	204 No Content – enrolment cancelled. 401 Unauthorized – authentication required. 404 Not Found – enrolment does not exist.
GET	/api/enrolments/{id}	Retrieves details of a specific enrolment.	Any	None	200 OK – enrolment returned. 404 Not Found – enrolment does not exist.
GET	/api/events/{eventId}/results	Retrieves all results for a specific event.	None	None	200 OK – results returned. 404 Not Found – event does not exist.
GET	/api/results/{id}	Retrieves a specific race result.	Any	None	200 OK – result returned. 404 Not Found – result does not exist.
POST	/api/enrolments/{enrolmentId}/result	Records the result for a participant's enrolment.	Organiser	{ position, time }	201 Created – result recorded. 400 Bad Request – invalid result data. 403 Forbidden – user is not an Organiser. 404 Not Found – enrolment does not exist. 409 Conflict – result already exists.

PUT	/api/results/{id}	Updates an existing race result.	Organiser	{ position, time }	200 OK – result updated. 400 Bad Request – invalid data. 404 Not Found – result does not exist.
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